- *Improved airframe and propulsion systems*: Use of lightweight composite materials and efficient electric or turbine engines increased endurance, payload capacity, and maneuverability.
- *Enhanced payload capabilities*: Integration of high-resolution cameras, thermal sensors, and reconnaissance modules for surveillance, mapping, and target acquisition.
- *Semi-autonomous operation*: Ability to execute partially automated flight missions with reduced operator input, marking the first step toward autonomous UAV systems.
- *Multi-domain applications*: Expanded use in military reconnaissance, environmental studies, meteorology, cartography, and agriculture, demonstrating drones' versatility beyond defense operations.

Figure 3.1 illustrates the basic characteristics of second-generation drones, highlighting their technological evolution from manually operated systems to semi-autonomous aerial platforms. The figure shows key components such as the GPS module, flight controller, radio receiver, and stabilization sensors (gyroscopes and IMUs) that enhance flight precision and stability. It also depicts the integration of digital telemetry and communication links between the drone and the ground control station (GCS), enabling real-time data exchange and mission monitoring. Advanced propulsion systems and lightweight composite airframes improved endurance and maneuverability, while the addition of high-resolution cameras and sensor payloads expanded functionality in reconnaissance, mapping, and environmental monitoring. Overall, the figure represents the defining features of GPS navigation, onboard electronics, and remote communication that transformed drones into reliable, multi-role aerial systems during the second-generation era.